



Book Introduction: Traffic Flow Dynamics Data, Models and Simulation

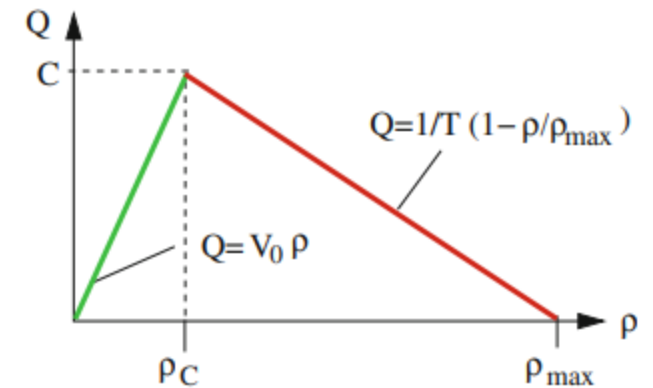
LWR Models with Triangular Fundamental Diagram



Triangular Fundamental Diagram

The simplest of the Lighthill–Whitham–Richards models uses a “triangular” fundamental diagram

$$Q_e(\rho) = \begin{cases} V_0 \rho & \text{if } \rho \leq \rho_c = \frac{1}{V_0 T + l_{\text{eff}}} \text{ (free traffic),} \\ \frac{1}{T} (1 - \rho l_{\text{eff}}) & \text{if } \rho_c < \rho \leq \rho_{\text{max}} = \frac{1}{l_{\text{eff}}} \text{ (congested traffic).} \end{cases}$$

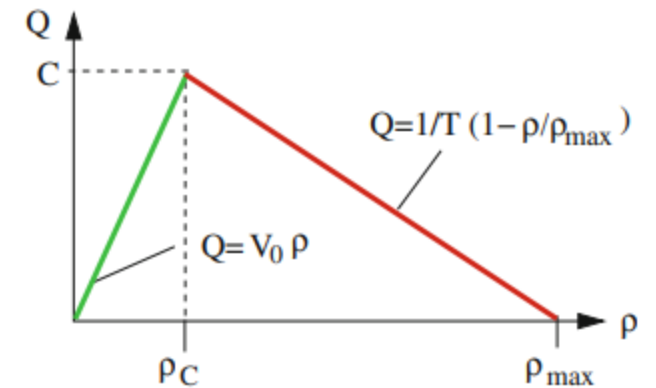


- V_0 : Free flow speed, represent driving a vehicle to its limits under unobstructed conditions.
- ρ_c : The critical point at which the road reaches its maximum traffic capacity (maximum flow rate Q_{max})
- T : The reaction delay time for a following vehicle to start moving after the vehicle ahead in stop-and-go traffic.
- l_{eff} : The space occupied by a single vehicle (vehicle length + the safety gap maintained at the front and rear when stationary). Therefore, the maximum jam density is $\rho_{\text{max}} = 1/l_{\text{eff}}$.

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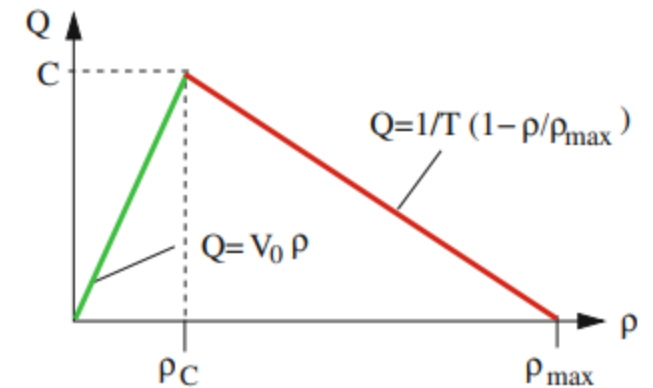
This model can be formulated in continuous and discrete variables:

1. The continuous version is called section-based model
2. The discrete version is formulated as an iterated coupled map with time and space discretized into time steps and cells, respectively, and supplemented by a special “supply-demand” update rule. This model is known as cell-transmission model (CTM)

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Section-based model:

Due to the specific properties of the triangular fundamental diagram, it is possible to break down the road into **road sections** defined such that each section has an **inhomogeneity** or **bottleneck** at the downstream end.

In each road section, there is at most one jam front.: it is sufficient to solve a **single integral** for the motion of the jam front.

$$\frac{\partial \rho}{\partial t} + \frac{dQ_e(\rho)}{d\rho} \frac{\partial \rho}{\partial x} = 0 \quad \text{LWR Model.}$$

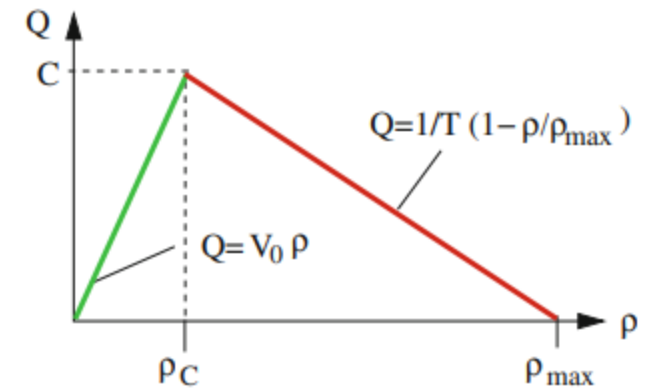
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The special shape of the fundamental diagram allows for efficient numeric update rules for the cell-transmission model (CTM) as well.

Another implication of the fixed propagation velocities is the absence of dispersion at the transitions from high to low density.



Model Parameters

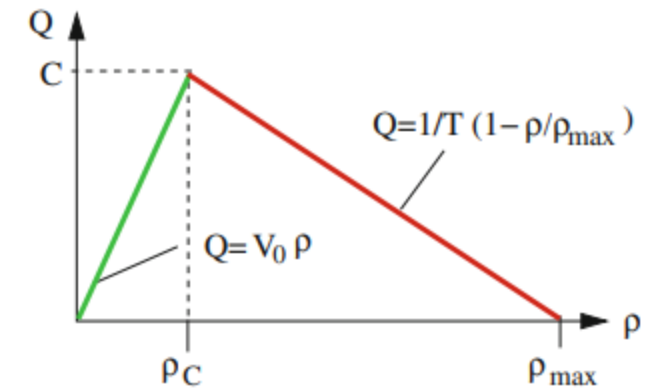
Table 8.1 Model parameters of the section-based model and the discrete cell-transmission model (CTM), and their typical values for highway, city, and pedestrian traffic

Parameter	Highway	City traffic	Pedestrian traffic
Desired speed V_0	110 km/h	50 km/h	1.3 m/s
Time gap T	1.4 s	1.2 s	0.5 ms
Maximum density $\rho_{\max}$	120 vehicles/km	120 vehicles/km	4 pedestrians/m ²

Since pedestrian flows are two-dimensional, the units of T and $\rho_{\max}$ and their meanings are different from those of vehicular traffic

As in the case in all macroscopic models, the parameters are averages over the individual vehicles (or pedestrians)

For vehicular traffic, the maximum density corresponds to the inverse of the minimum distance headway l_{eff}

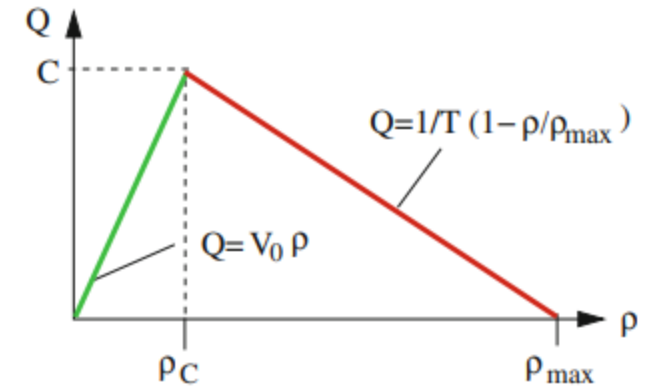


$$l_{\text{eff}} = s_0 + l = \frac{1}{\rho_{\max}} .$$

Characteristic Properties

Analytical inverse function of the fundamental diagram.

In real-world , the most important input and control variable (i.e., open) systems is the **traffic flow**, while **local speed** and **density** are dependent variables of the flow and the traffic state.



However, this function is not unique since, for a given flow Q , there are two possible density values: one from the so-called branch for free flow, $\rho_{free}(Q)$ and one from the branch for congested traffic, $\rho_{cong}(Q)$

$$\rho_{free}(Q) = \frac{Q}{V_0},$$

$$\rho_{cong}(Q) = \frac{1 - QT}{l_{eff}} = \rho_{\max}(1 - QT).$$

$$Q_e(\rho) = \begin{cases} V_0 \rho & \text{if } \rho \leq \rho_C = \frac{1}{V_0 T + l_{eff}} \quad (\text{free traffic}), \\ \frac{1}{T} (1 - \rho l_{eff}) & \text{if } \rho_C < \rho \leq \rho_{\max} = \frac{1}{l_{eff}} \quad (\text{congested traffic}). \end{cases}$$

Characteristic Properties

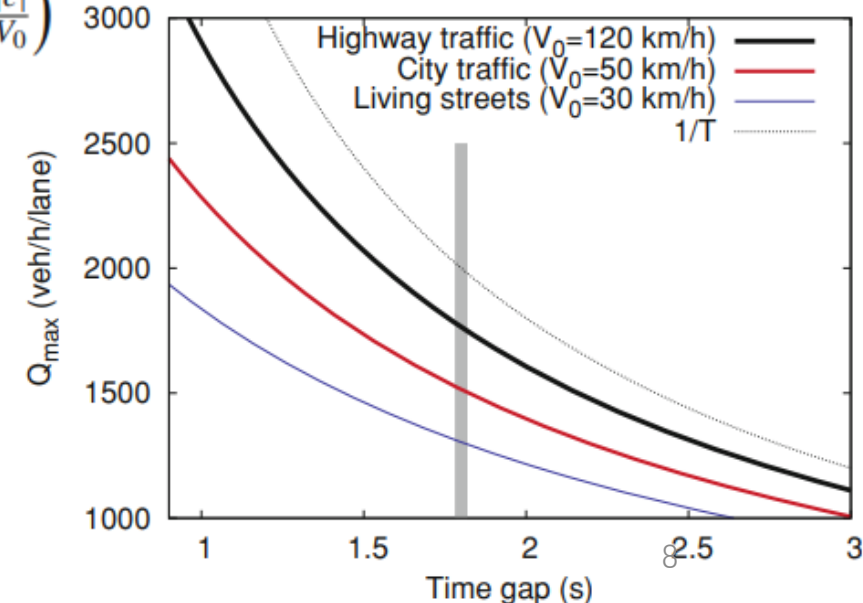
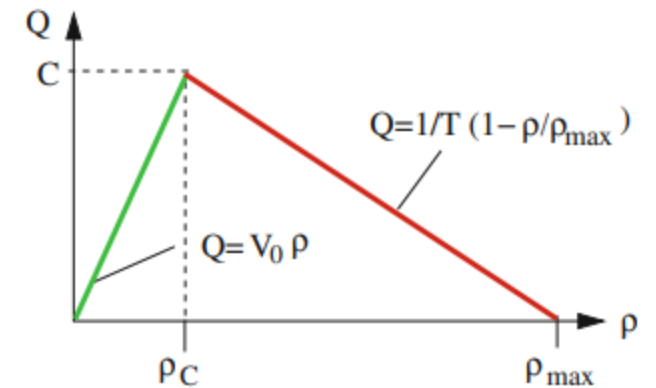
Analytical inverse function of the fundamental diagram.

The maximum flow Q_{max} , i.e., the effective capacity C **per lane** on a homogeneous road section is given by the intersection of the two branches.

$$\left\{ \begin{array}{l} \rho_{free}(Q) = \frac{Q}{V_0}, \\ \rho_{cong}(Q) = \frac{1 - QT}{l_{eff}} = \rho_{max}(1 - QT). \end{array} \right. \Rightarrow \frac{C}{I} = Q_{max} = \frac{1}{T + \frac{l_{eff}}{V_0}} = \frac{1}{T \left(1 + \frac{|c|}{V_0}\right)}$$

$$|c| = \frac{l_{eff}}{T}$$

The capacity is always smaller than the inverse of the mean time gap T



Characteristic Properties

Constant propagation velocity of density variations in congested traffic.

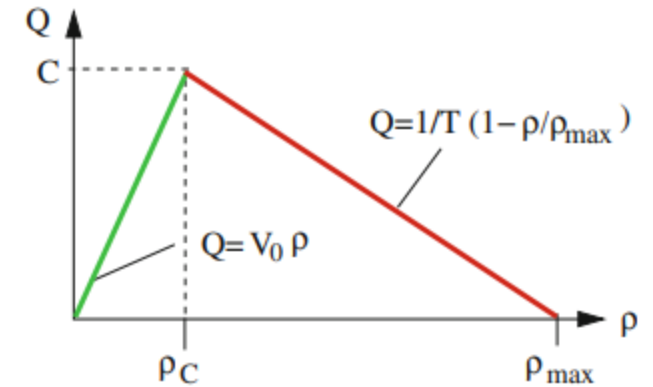
In congested traffic, the propagation velocities of flow, density, or speed variations are given by the falling slope of the fundamental diagram:

$$c_{\text{cong}} = c = \left. \frac{dQ_e}{d\rho} \right|_{\rho > \rho_C} = -\frac{l_{\text{eff}}}{T} = -\frac{1}{\rho_{\text{max}} T}.$$

The negative value of c means that speed and density **variations propagate backwards**, not only relative to the other vehicles but also for a stationary observer

The main influencing factors of the propagation velocity:
Perturbations propagate by one effective vehicle **length** per **time gap**

The absolute value $|c|$ is about 14–16 km/h in Europe, and 18–20 km/h in the United States.



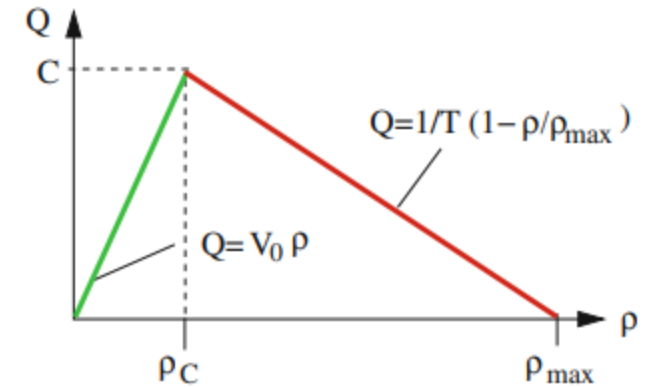
Characteristic Properties

Propagation velocities of transitions from free to congested traffic

Let us denote the **free traffic** state **upstream** of the transition by the index 1 and the **congested traffic** state by index 2.
Then, the upstream jam front propagates with the velocity

$$c_{up} = c_{12} = \frac{Q_2 - Q_1}{\rho_2 - \rho_1} = \frac{Q_2 - Q_1}{\rho_{max}(1 - Q_2 T) - Q_1 / V_0}.$$

This velocity can take on any value between c and V_0 and thus the front may move in either direction (growing or shrinking traffic jams).



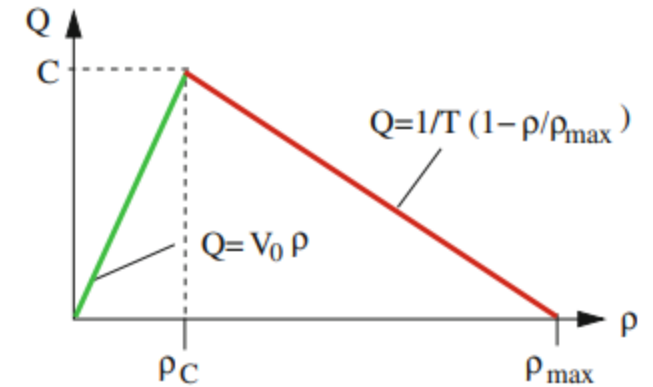
Characteristic Properties

Constant propagation velocity of downstream jam fronts

The outflow at the downstream end of a moving traffic jam is characterized by the maximum-flow state (ρ_c, Q_{max}) . For a triangular fundamental diagram, the maximum-flow state corresponds to the top of the triangle, so the **downstream front propagates at the same velocity** as small variations within the jam

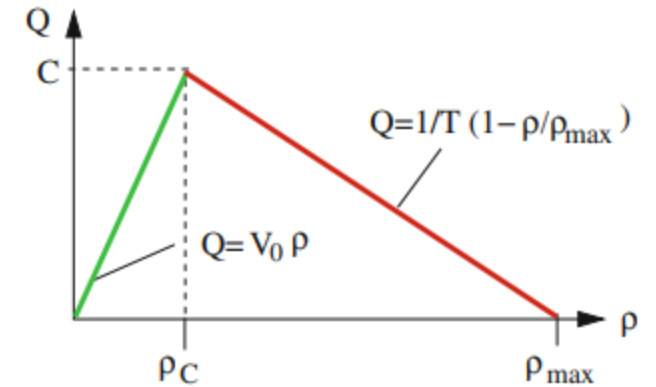
$$c_{down} = c = -\frac{1}{\rho_{max} T} = -\frac{l_{eff}}{T}.$$

If there is a bottleneck, the downstream front gets stationary and pinned at the bottleneck, and the outflow is reduced to the capacity of the bottleneck.



Model Formulation with Measurable Quantities

In Equation, we formulated the triangular fundamental diagram in terms of parameters directly describing properties of the vehicles and aspects of driving behavior: Mean desired speed V_0 in free traffic, mean desired time gap T in car-following situations, and mean effective vehicle length $l_{eff} = 1/\rho_{max}$.



$$Q_e(\rho) = \begin{cases} V_0 \rho & \text{if } \rho \leq \rho_C = \frac{1}{V_0 T + l_{eff}} \quad (\text{free traffic}), \\ \frac{1}{T} (1 - \rho l_{eff}) & \text{if } \rho_C < \rho \leq \rho_{max} = \frac{1}{l_{eff}} \quad (\text{congested traffic}). \end{cases}$$

We cannot directly determine the time gap and the effective length with the most commonly available aggregated stationary detector data

Model Formulation with Measurable Quantities

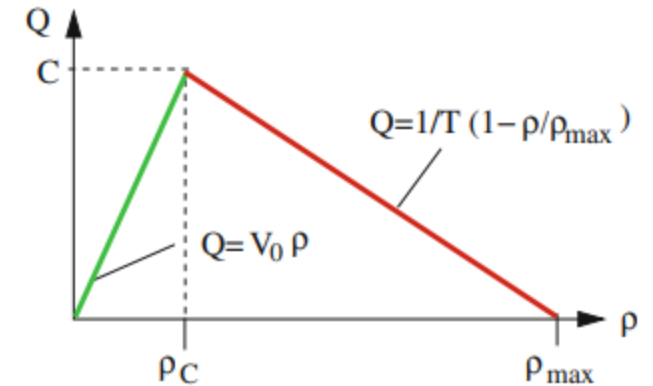
$$c_{\text{cong}} = c = \left. \frac{dQ_e}{d\rho} \right|_{\rho > \rho_C} = -\frac{l_{\text{eff}}}{T} = -\frac{1}{\rho_{\text{max}} T}.$$

$$\frac{C}{I} = Q_{\text{max}} = \frac{1}{T + \frac{l_{\text{eff}}}{V_0}} = \frac{1}{T \left(1 + \frac{|c|}{V_0}\right)}$$

Express T and l_{eff} by the more easily accessible quantities effective capacity Q_{max} per lane, and propagation velocity c in congested traffic:

$$\frac{1}{l_{\text{eff}}} = \rho_{\text{max}} = Q_{\text{max}} \left(\frac{1}{V_0} - \frac{1}{c} \right), \quad T = \frac{1}{Q_{\text{max}} \left(1 - \frac{c}{V_0} \right)}.$$

$$Q_e(\rho) = \begin{cases} V_0 \rho & \text{if } \rho \leq \frac{Q_{\text{max}}}{V_0} \text{ (free flow),} \\ Q_{\text{max}} \left[1 - \frac{c}{V_0} \right] + c\rho & \text{if } \rho > \frac{Q_{\text{max}}}{V_0} \text{ (congested flow).} \end{cases}$$



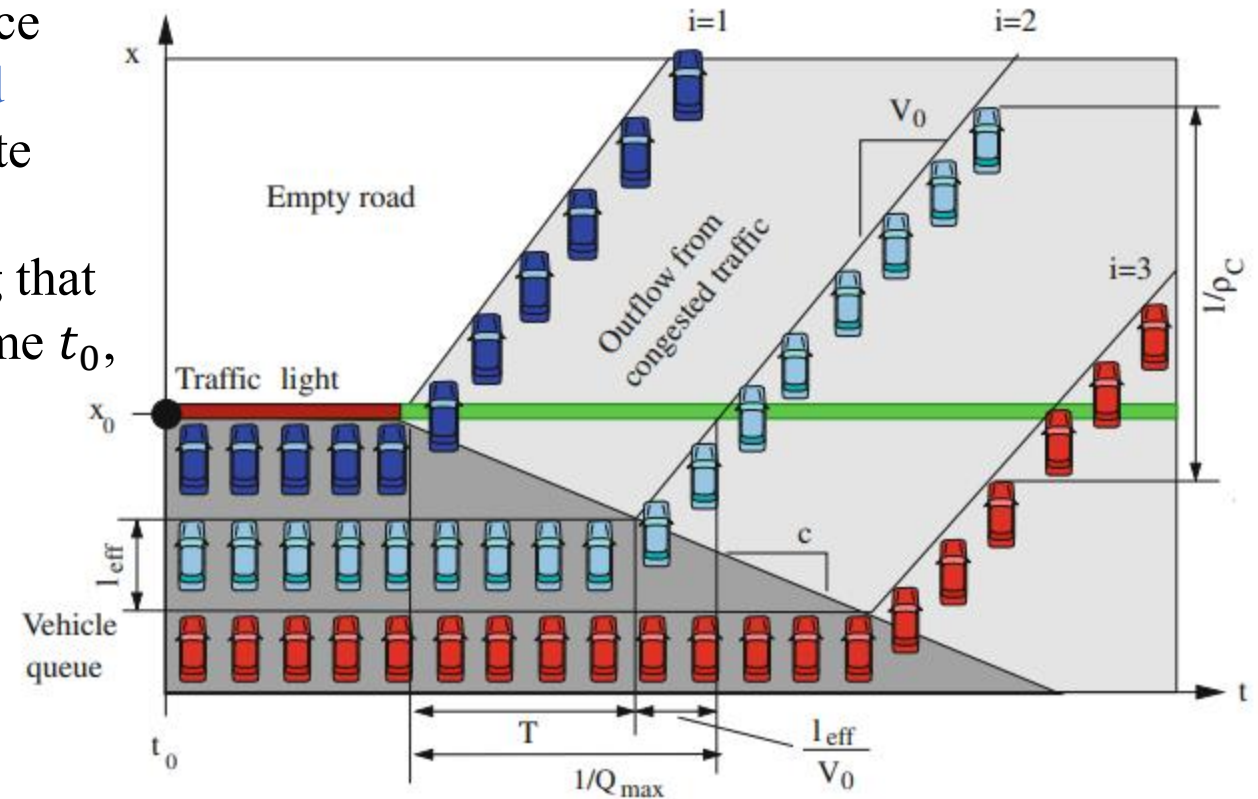
- We can approximatively estimate Q_{max} by the outflow of moving traffic waves.
- To estimate the propagation velocity c , we determine the time interval which is needed by the downstream front of a moving traffic wave or another identifiable structure to pass two consecutive stationary detectors.

Relation to Car-Following Models

By using the definitions of the **density** (inverse distance headway between two vehicles) and the **vehicle speed** (slope of the tangent of the trajectory), we can generate **trajectories** $x_i(t)$ of individual vehicles reflecting the macroscopic density and local speed fields. Assuming that the trajectory of vehicle i crosses the location x_0 at time t_0 , the trajectories are defined by

$$x_i(t_0) = x_0, \quad \int_{x_{i+1}(t_0)}^{x_i(t_0)} \rho(x, t_0) dx = 1, \quad v_i(t) = \frac{dx_i}{dt} = V(x_i(t)),$$

Providing a single point (x_0, t_0) crossed by a trajectory defines all trajectories for all locations at all times (including the past) within the spatiotemporal interval of the macroscopic data.

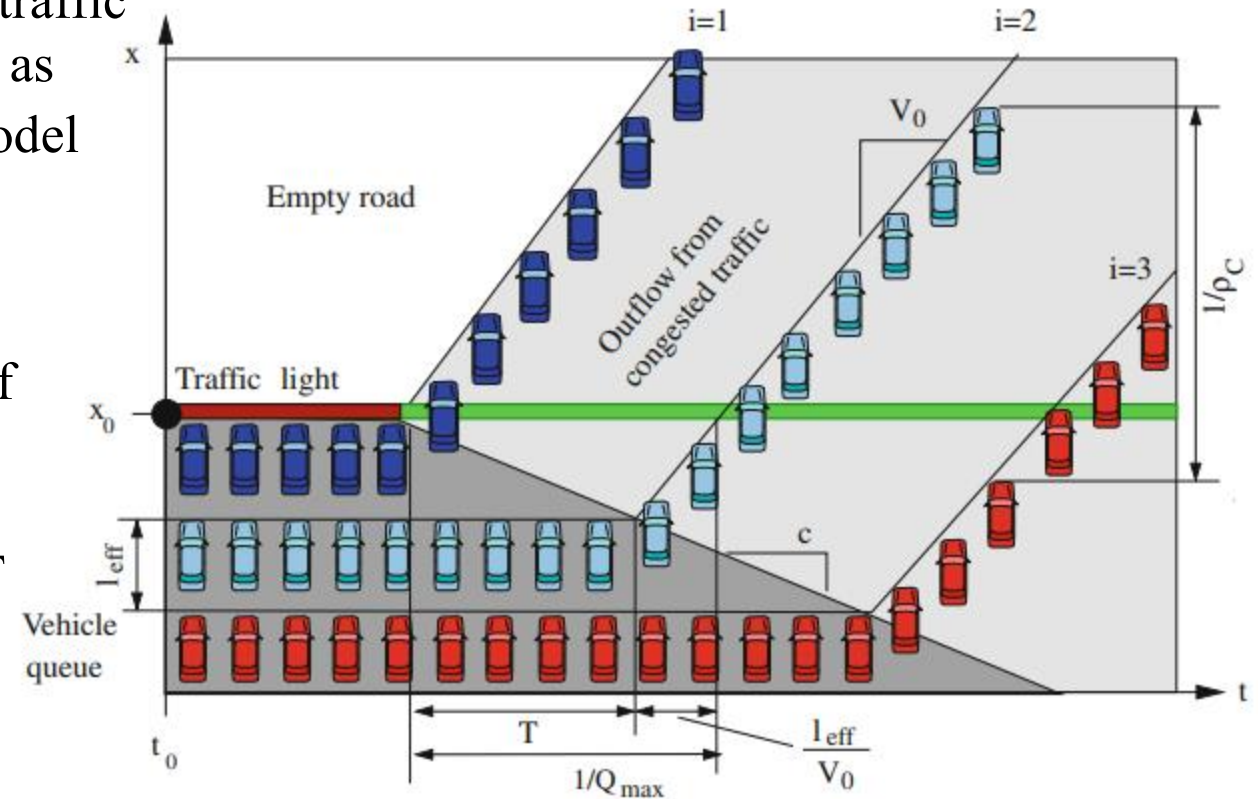


Relation to Car-Following Models

In the case of the section-based model and congested traffic ($\rho(x,t) \geq \rho_C$), these generated trajectories are the same as those produced by Newell's model, a car-following model

$$v_i(t + T) = v_{i-1}(t).$$

This equation describes a behavior where the driver of vehicle i exactly copies the speed profile of his or her leader $i - 1$ at a constant delay T which is equal to the local time gap. In this sense, the time gap parameter T may be interpreted as a reaction time.



However, this is only true within congested traffic or for the maximum-flow state.

Relation to Car-Following Models

In congested traffic, the section-based model describes density and speed variations propagating with constant velocity c , so $V(x, t) = V(x - ct, 0)$ for all times t

The speed profile of the leading vehicle $i - 1$ is given by the local speed field $V(x, t)$ along the trajectory

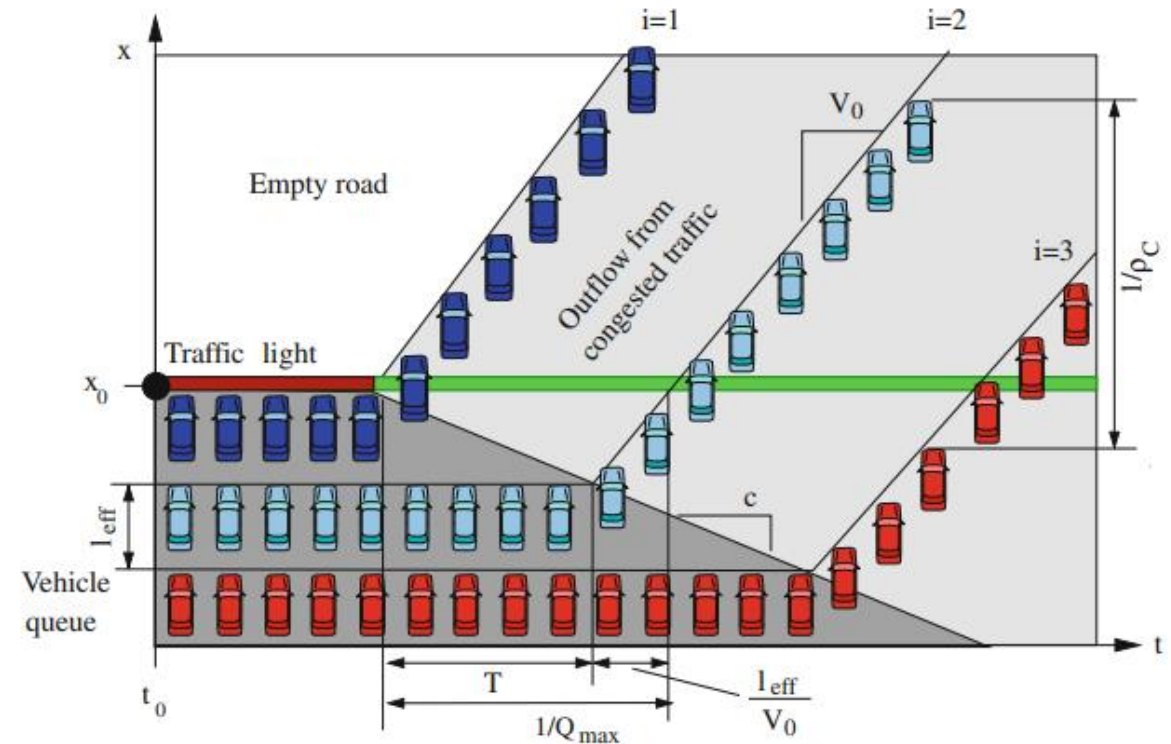
$$v_{i-1}(t) = V(x_{i-1}(t), t) = V(x_{i-1}(t) - ct, 0).$$

Furthermore, by means of the triangular fundamental diagram, the time gap of vehicle i to the leading vehicle $i - 1$ is given by T for congested conditions ($\rho > \rho_C$) as required here.

Thus, the position of vehicle i at time t is

$$x_i(t) = x_{i-1}(t) - v_{i-1}T - l_{eff} = x_{i-1}(t - T) - l_{eff}$$

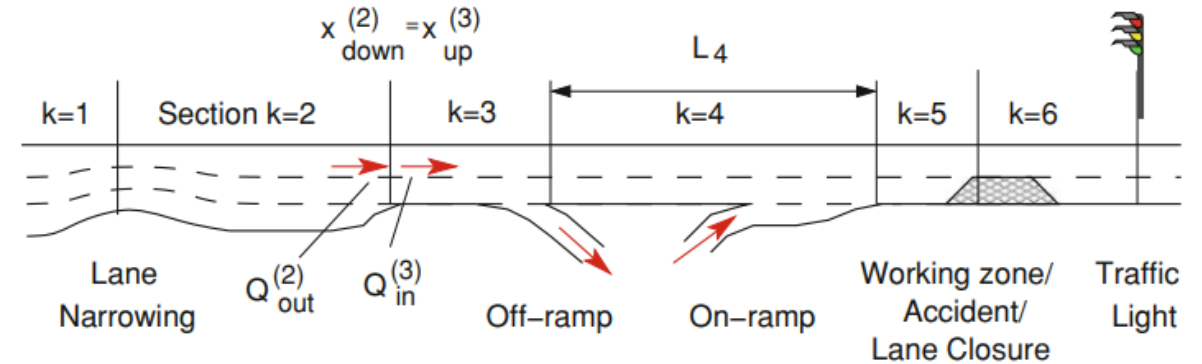
for any time.



$$\begin{aligned} v_i(t) &= V(x_i(t), t) = V(x_i(t) - ct, 0) \\ &= V(x_{i-1}(t - T) - l_{eff} - ct, 0) \\ &= V\left(x_{i-1}(t - T) - c\left(t + \frac{l_{eff}}{c}\right), 0\right) \\ &= V(x_{i-1}(t - T) - c(t - T), 0) = V(x_{i-1}(t - T), t - T) \\ &= v_{i-1}(t - T), \end{aligned}$$

Definition of Road Sections

When passing flow-conserving bottlenecks all vehicles pass from road section k to $k + 1$, no vehicle leaves section k or enters section $k + 1$ differently. From the viewpoint of an economist, flow-conserving bottlenecks represent a change in the supply (local road capacity).



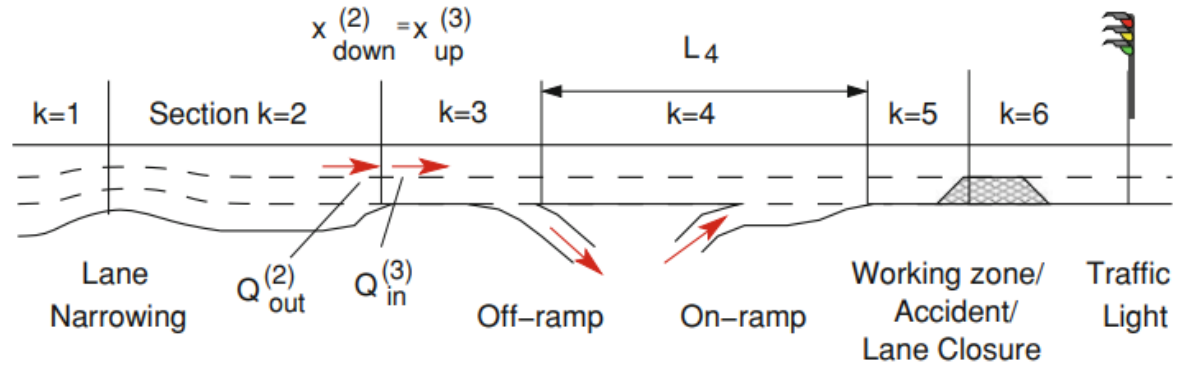
Non-flow-conserving bottlenecks contain sources and sinks constituted by on-ramps, off-ramps, junctions, interchanges, intersections, and similar. From the viewpoint of an economist, non-flow-conserving bottlenecks represent a change in the traffic demand.

We can also distinguish the bottlenecks whether they are permanent on the time scales of typical simulations

Definition of Road Sections

Regardless of the type of bottleneck, the influence on traffic flow can be characterized by a single property, at least within the class of LWR models: The bottleneck strength.

In LWR or other first-order macroscopic traffic flow models, bottlenecks are characterized by the reduction of local capacity with respect to the homogeneous upstream road section. Only a single property is relevant for traffic flow dynamics: The effective bottleneck strength $\Delta Q = \frac{\Delta C}{I}$, i.e., the capacity reduction $\Delta C = C_{\text{upstream}} - C_{\text{bottl}}$ divided by the number I of lanes in this section.



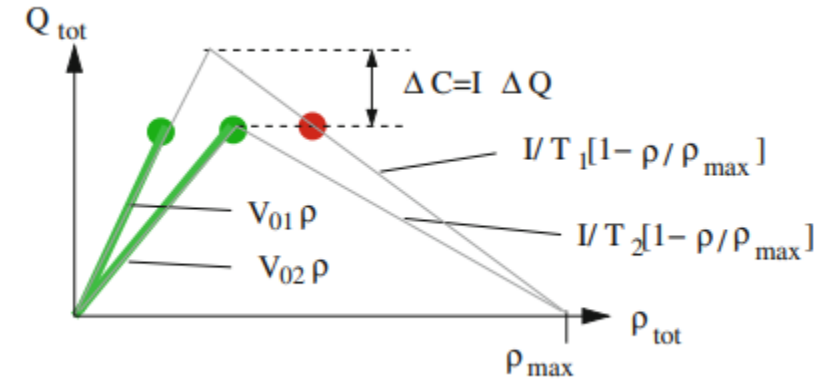
Definition of Road Sections

Classical Flow-Conserving Bottlenecks

This includes all flow-conserving bottlenecks that are caused by local changes of the road attributes except changes in the number of lanes.

The defining property of this type of bottleneck is a local reduction ΔQ of the capacity per lane. In all microscopic and macroscopic traffic flow models, this type of flow-conserving bottleneck is modeled by local changes of the model parameters, i.e., by changing the average driving behavior.

Changing the mean desired speed V and/or the time gap T while the average vehicle length $l_{eff} = 1/\rho_{max}$ should remain unchanged.



$$\Delta Q_{cl} = \frac{\Delta C_{cl}}{I} = \frac{V_{01}}{V_{01}T_1 + l_{eff}} - \frac{V_{02}}{V_{02}T_2 + l_{eff}}$$

Definition of Road Sections

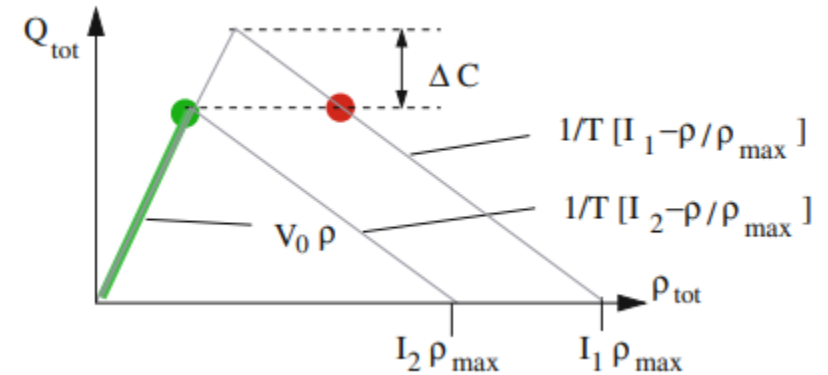
Lane-Drop Bottlenecks

This type of bottleneck plays an intermediate role. From a global viewpoint over all lanes, it is a flow-conserving bottleneck

The fundamental diagram for the total flow and density will be congruently inflated or deflated whenever the number I of lanes changes. Particularly, when reducing the number of lanes from I_1 to $I_2 < I_1$, the total and effective capacities are reduced by

$$\Delta C_{\text{lane drop}} = (I_1 - I_2) Q_{\max}, \quad \Delta Q_{\text{lane drop}} = \frac{I_1 - I_2}{I_1} Q_{\max},$$

Particularly, the LWR models will not produce any congestion inside the section with a reduced number of lanes



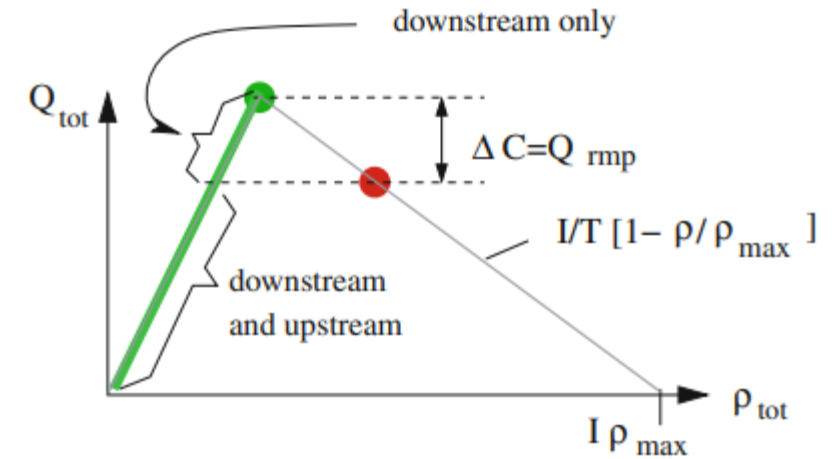
Definition of Road Sections

On-Ramps

on-ramp bottlenecks are caused by the increasing demand of the merging vehicles (additional flow Q_{rmp}) rather than by a decreasing supply (local capacity reduction by ΔC)

If we assume that all on-ramp vehicles can merge to the main-road regardless of traffic conditions, the main-road traffic flow upstream of the on-ramp is restricted by the capacity C of all the main-road lanes at the location of the on-ramp minus the ramp flow Q_{rmp}

$$\Delta C_{\text{rmp}} = Q_{\text{rmp}}, \quad \Delta Q_{\text{rmp}} = \frac{Q_{\text{rmp}}}{I}$$



If the on-ramp consists of more than one lane, the on-ramp flow is to be understood as the total flow over all ramp lanes.